

20250609EB_MN (NCT, GCamp, ER-GEM)

Project: Yi-Shuan Tseng Holt Lab

Author: Yi Shuan Tseng

Status: In progress

Entry Created On: 26 Jun 2025 18:15:54 UTC

Entry Last Modified: 03 Jul 2025 20:11:47 UTC

Export Generated On: 01 Sep 2026 18:26:59 UTC

2025年6月9日 星期一

Differentiation: BP w/ 1.5mM pyruvate.

Experiment: NB-A with 1mM pyruvate, N2, B27, Glutamax, T8 supps

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年6月11日 星期三

T2 -> T3

2025年6月13日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年6月16日 星期一

T4 -> T5

2025年6月18日 星期三

T5 -> T6

2025年6月20日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年6月23日 星期一

T6 -> T7

Dissociation (performed by Stephanie)

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) o/n on
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatant (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

NOTE

- Use newly prepared 0.25% trypsin with DNase from last week
- The trypsin is very strong and the cells clump and became viscous within 5 min

coating date			^
	A	B	
1	0.01% PLO with Na borate	2025/6/13	
2	laminin	2025/6/14	

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P16+18	6	

Table1			^
	A	B	
1	run1	volume (mL)	
2	trypsin	3	
3	HI-FBS/CTWM	6	

Table3								
	A	B	C	D	E	F	G	H
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells
2	0.623	96	5	3.115	A	24	222.5	14

Well1							^
	1	2	3	4	5	6	
A		74 5uL/w ell	129 5uL/w ell				
B							
C				110 3uL/w ell	111 3uL/w ell		
D							

- 2025年6月24日 星期二
- Change T7 -> T7 if added virus
- 2025年6月26日 星期四
- T7 -> BP T8
- 2025年6月28日 星期六
- 1/2 T8 -> 1/2 T8
- 2025年6月30日 星期一
- 1/2 T8 -> 1/2 T8
- 2025年7月2日 星期三
- 1/2 T8 -> 1/2 T8 for Stephanie's wells.
- Stephanie treated vLH129 with 1ug/mL Dox at around 1:30pm.

Imaging (NCT)

Try the condition 7 vs. 8 from 20250527MN

condition from 20250527MN														/
	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	condition	media	pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume	
2	7	NB-A	1mM	v	20	40	v	20	v	2.5	4.55		2000	
3	8	NB-A	1mM	v	20	40	v	20	v	0			2000	

Prepare 20 mL 0mM media first:

condition from 20250527MN1														/
	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	media	pyruvate (100x)	100mM pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume (uL)	
2	NB-A	1mM	200	v	200	400	v	200	v	0			20000	

Aliquot 2mL for each condition, and add glucose:

NCT treat time										
	A	B	C	D	E	F	G	H	I	J
1		110			111					
2	condition#	treat time	image time	treat time	image time	glucose conc	diluted from (mM)	glucose volume	total volume (uL)	SirDNA (1:5000)
3	1	13:10	14:10	10:00	11:00	2.5	1100	4.55	2000	0.4
4	2	13:55	14:55	10:30	11:30	0.15	2.5	120.00	2000	0.4
5	3	14:30	15:30	11:15	12:15	0		0	2000	0.4

Well2						
	1	2	3	4	5	6
A		74 5uL/w ell	129 5uL/w ell			
B				1	1	
C				2	2	
D				3	3	

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

10 positions/well (10 positions would be the max. there is only 2 sec between loops)

Used 16 bit instead of 12 bit (the S/N is much better!!!!)

110: pre-activation: 3 loops, 30s interval; activation: 30 loops, 30s interval

111: pre-acitvation: 3 loops, 30s interval; activation: **25 loops**, 30s interval

*Note: 110_6 position 2 1st frame is out of focus (a bubble passed by). 2nd frame is a bit affected. --> exclude it.